

BAULKHAM HILLS HIGH SCHOOL MARKING COVER SHEET



2011 YEAR 12 JUNE ASSESSMENT TASK EXTENSION 2

STUDENT NUMBER: _____

TEACHERS NAME: _____

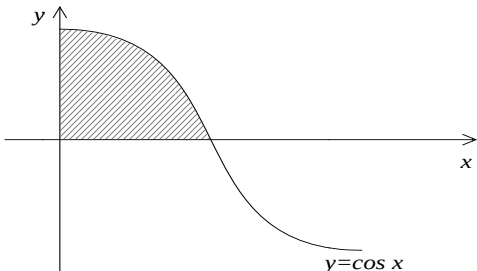
QUESTION	MARK
1	
2	
3	
4	
TOTAL	/
PERCENTAGE	%

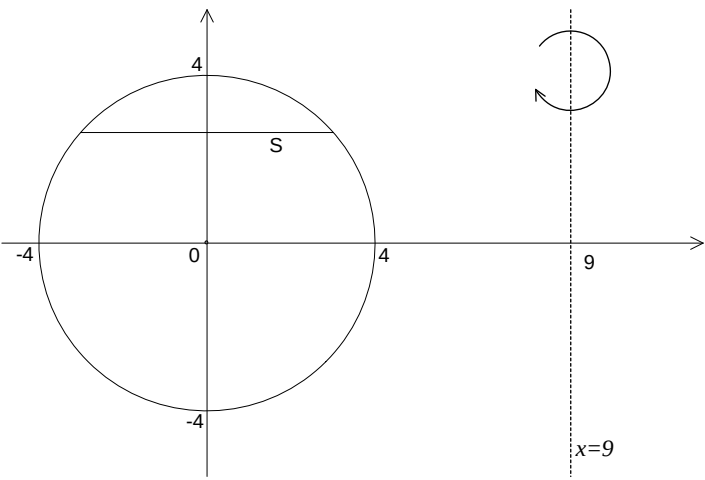
Topics Tested: Integration and Volumes



YEAR 12 EXTENSION 2 MATHEMATICS ASSESSMENT JUNE 2011

TIME : 70 MINUTES

NAME		RESULT	
DIRECTIONS ▪ Full working should be shown in every question. Marks may be deducted for careless or badly arranged work. ▪ Use black or blue pen only (<i>not pencils</i>) to write your solutions. ▪ No liquid paper or correction tape is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.			
Question 1 (10 marks)			Marks
a)	Find the indefinite integrals:		
i)	$\int \frac{dx}{\sqrt{4-9x^2}}$	2	
ii)	$\int \frac{dx}{x^2+2x+5}$	3	
iii)	$\int \sin^{-1} x \, dx$	2	
b)	Evaluate: $\int_0^2 \frac{x^2}{x^6+64} \, dx$	3	
Question 2 (11 marks) - Start a new page			
a)	Use the method of <u>cylindrical shells</u> to find the volume of the solid generated when shaded region is rotated about the line $x = \pi$.	4	
			
b) i)	Show that $\int_0^1 \frac{1}{(5x+3)(x+1)} \, dx = \frac{1}{2} \ln \frac{4}{3}$	3	
ii)	Hence find $\int_0^{\frac{\pi}{2}} \frac{1}{4 \sin x - \cos x + 4} \, dx$ using the substitution $t = \tan \frac{x}{2}$	4	

Question 3 (12 marks) - Start a new page		Marks
a)	The circle $x^2 + y^2 = 16$ is rotated about the line $x = 9$ to form a ring (i.e. a torus). When the circle is rotated, the line segment S at height y sweeps out an annulus. 	
i)	Show that the area of the annulus is equal to $36\pi\sqrt{16 - y^2}$.	3
ii)	Hence find the volume of the ring.	3
b)	Given $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$, i) Show that $I_n + I_{n-2} = \frac{1}{n-1}$ where n is an integer and $n \geq 3$	4
	ii) Hence evaluate I_7 .	2
Question 4 (11 marks) - Start a new page		Marks
a)	A certain solid has a circular base of radius 4 units. The centre of the base is the origin. The cross-sections at right-angles to the x-axis are isosceles triangles. The height h of each of these triangles is given by $h = 16 - x^2$.	
i)	Show that the volume of the solid is given by $V = 2 \int_0^4 (16 - x^2)^{\frac{3}{2}} \, dx$.	2
ii)	Hence find the volume V	4
b)	i) Show that $\int_0^{2a} f(x) \, dx = \int_0^a [f(x) + f(2a - x)] \, dx$	3
	ii) Hence find the value of the constant k given that $\int_0^{\pi} \frac{x}{2 + \sin x} \, dx = k \int_0^{\pi/2} \frac{1}{2 + \sin x} \, dx$	2
-- End of Exam --		

Ext 2 June Task 2011 SOLUTIONS.

Question 1.

a) i) $\int \frac{dx}{\sqrt{4-9x^2}}$ $\begin{cases} u=3x \\ du=3 \cdot dx \end{cases}$

$$= \int \frac{\frac{1}{3} \cdot du}{\sqrt{4-u^2}}$$

$$= \frac{1}{3} \sin^{-1} \frac{u}{2} + c$$

$$= \frac{1}{3} \sin^{-1} \frac{3x}{2} + c \quad \begin{matrix} \sin^{-1} = 1 \\ \text{The rest} = 1 \end{matrix}$$

ii) $\int \frac{dx}{x^2+2x+5}$

$$= \int \frac{dx}{(x+1)^2+4}$$

$$= \frac{1}{2} \tan^{-1} \frac{x+1}{2} + c$$

iii) $\int \sin^{-1} x \cdot 1 \cdot dx$

$$= uv - \int v u' \cdot dx \quad \begin{cases} u = \sin^{-1} x & u' = \frac{1}{\sqrt{1-x^2}} \\ v = x & v' = 1 \end{cases}$$

$$= x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} \cdot dx$$

$$= x \sin^{-1} x - \frac{1}{2} \int \frac{2x}{\sqrt{1-x^2}} \cdot dx$$

$$= x \sin^{-1} x + \frac{1}{2} \int \frac{du}{\sqrt{u}} \quad \begin{cases} u=1-x^2 \\ du=-2x \cdot dx \end{cases}$$

$$= x \sin^{-1} x + \frac{1}{2} \cdot 2u^{1/2} + c$$

$$= x \sin^{-1} x + \sqrt{1-x^2} + c$$

$\swarrow$ $\swarrow$
 $\leftarrow$ $\leftarrow$

b) $\int_0^2 \frac{x^2}{x^6+64} \cdot dx$

$$\begin{cases} u=x^3 \\ du=3x^2 \cdot dx \\ \text{If } x=0, u=0 \\ \text{If } x=2, u=8 \end{cases}$$

$$= \int_0^8 \frac{\frac{1}{3} \cdot du}{u^2+64}$$

$$= \frac{1}{3} \left[\frac{1}{8} \tan^{-1} \frac{u}{8} \right]_0^8$$

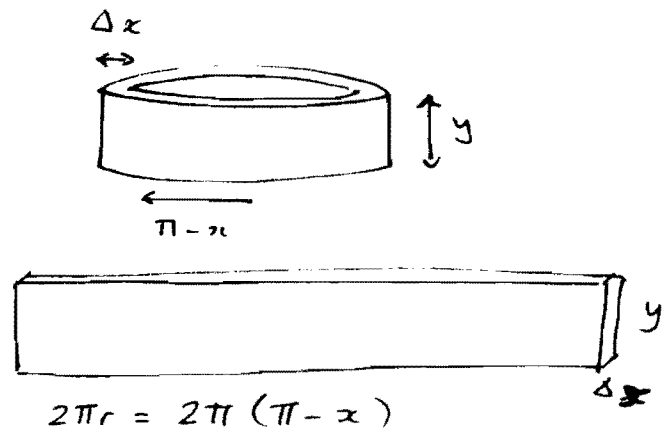
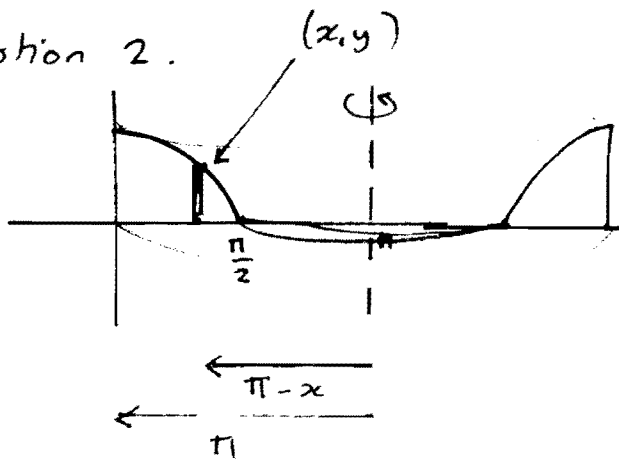
$$= \frac{1}{24} (\tan^{-1} 1 - \tan^{-1} 0)$$

$$= \frac{1}{24} \times \frac{\pi}{4}$$

$$= \frac{\pi}{96}$$

Question 2.

a)



$$\Delta V = 2\pi (\pi - x) \cdot y \cdot \Delta x$$

$$= 2\pi (\pi - x) \cdot \cos x \cdot \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{\pi/2} 2\pi (\pi - x) \cos x \cdot \Delta x$$

$$= 2\pi \int_0^{\pi/2} \pi \cos x - x \cos x \cdot dx$$

$$= 2\pi \left[\pi \sin x \right]_0^{\pi/2} - 2\pi \underbrace{\int_0^{\pi/2} x \cdot \cos x \cdot dx}_I$$

$$= 2\pi (\pi - 0) - 2\pi \cdot I$$

$$= 2\pi^2 - 2\pi I$$

Now

$$I = \int_0^{\pi/2} x \cos x \cdot dx$$

$$u = x \quad u' = 1$$

$$v = \sin x \quad v' = \cos x$$

$$= \left[x \sin x \right]_0^{\pi/2} - \int_0^{\pi/2} \sin x \cdot dx$$

$$= \left(\frac{\pi}{2} - 0 \right) + \left[\cos x \right]_0^{\pi/2}$$

$$= \frac{\pi}{2} + \cos \frac{\pi}{2} - \cos 0$$

$$= \frac{\pi}{2} + 0 - 1$$

$$= \frac{\pi}{2} - 1$$

$$\therefore V = 2\pi^2 - 2\pi \left(\frac{\pi}{2} - 1 \right)$$

$$= 2\pi^2 - \pi^2 + 2\pi$$

$$= \pi^2 + 2\pi \quad \text{units}^3$$

$$b) i) \frac{1}{(5x+3)(x+1)} = \frac{A}{5x+3} + \frac{B}{x+1}$$

$$1 = A(x+1) + B(5x+3)$$

$$\text{If } x = -1, \quad 1 = 0 + B(-2) \quad \therefore B = -\frac{1}{2}$$

$$\text{If } x = -\frac{3}{5}, \quad 1 = A\left(\frac{2}{5}\right) + 0 \quad \therefore A = \frac{5}{2}$$

$$\therefore \int_0^1 \frac{1}{(5x+3)(x+1)} \cdot dx$$

$$= \int_0^1 \frac{5/2}{5x+3} - \frac{1/2}{x+1} \cdot dx$$

$$= \left[\frac{1}{2} \ln(5x+3) - \frac{1}{2} \ln(x+1) \right]_0^1$$

$$= \left(\frac{1}{2} \ln 8 - \frac{1}{2} \ln 2 \right) - \left(\frac{1}{2} \ln 3 - \frac{1}{2} \ln 1 \right)$$

$$= \frac{1}{2} \ln 4 - \frac{1}{2} \ln 3$$

$$= \frac{1}{2} \ln \frac{4}{3}$$

$$(ii) \int_0^{\pi/2} \frac{1}{4 \sin x - \cos x + 4} \cdot dx$$

$$= \int_0^1 \frac{1}{\frac{8t}{1+t^2} - \frac{1-t^2}{1+t^2} + 4} \cdot \frac{2}{1+t^2} \cdot dt$$

$$t = \tan \frac{x}{2}$$

$$2 \cdot dt = \frac{1}{2} \sec^2 \frac{x}{2} \cdot dx$$

$$\frac{2 \cdot dt}{1+t^2} = \frac{1}{1+t^2} \cdot dx$$

$$\text{If } x = 0, \quad t = 0$$

$$\text{If } x = \pi/2, \quad t = 1$$

$$= \int_0^1 \frac{2}{8t - 1 + t^2 + 4 + 4t^2} dt \quad |$$

$$= \int_0^1 \frac{2}{5t^2 + 8t + 3} \cdot dt \quad |$$

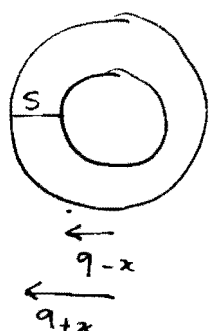
$$= 2 \int_0^1 \frac{1}{(5t+3)(t+1)} \cdot dt$$

$$= 2 \left(\frac{1}{2} \ln \frac{4}{3} \right) \quad \text{from result in (i)} \quad |$$

$$= \underline{\ln \frac{4}{3}}$$

Question 3.

a) i) Endpts of S are (x, y) and $(-x, y)$



$$A = \pi \left((9+x)^2 - (9-x)^2 \right) \quad |$$

$$= \pi \cdot 36x$$

$$= 36\pi \cdot \sqrt{16-y^2}$$

$$\left\{ \begin{array}{l} \text{but } x^2 + y^2 = 16 \\ \therefore x = \sqrt{16-y^2} \end{array} \right. \quad |$$

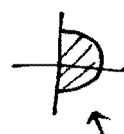


$$\Delta V = 36\pi \cdot \sqrt{16-y^2} \cdot \Delta y$$

$$V = 36\pi \int_{-4}^4 \sqrt{16-y^2} \cdot dy \quad |$$

$$= 36\pi \times \frac{1}{2} \pi (4)^2 \quad |$$

$$= 288\pi^2 \text{ units}^3 \quad |$$



(area corresponding to integral)

$$\begin{aligned}
 b) i) I_n &= \int_0^{\pi/4} \tan^n x \cdot dx \\
 &= \int_0^{\pi/4} \tan^{n-2} x \cdot \tan^2 x \cdot dx \\
 &= \int_0^{\pi/4} \tan^{n-2} x \cdot (\sec^2 x - 1) \cdot dx \\
 &= \int_0^{\pi/4} \tan^{n-2} x \cdot \sec^2 x \cdot dx - \int_0^{\pi/4} \tan^{n-2} x \cdot dx \\
 &= \left[\frac{\tan^{n-1} x}{n-1} \right]_0^{\pi/4} - I_{n-2} \\
 &= \left(\frac{1}{n-1} - 0 \right) - I_{n-2}
 \end{aligned}$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1}$$

$$ii) I_7 + I_5 = \frac{1}{6}$$

$$I_5 + I_3 = \frac{1}{4}$$

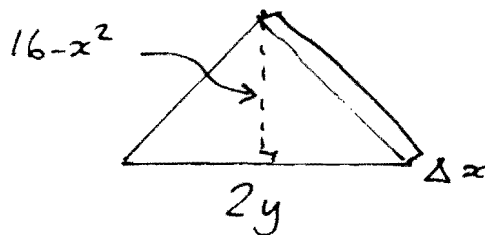
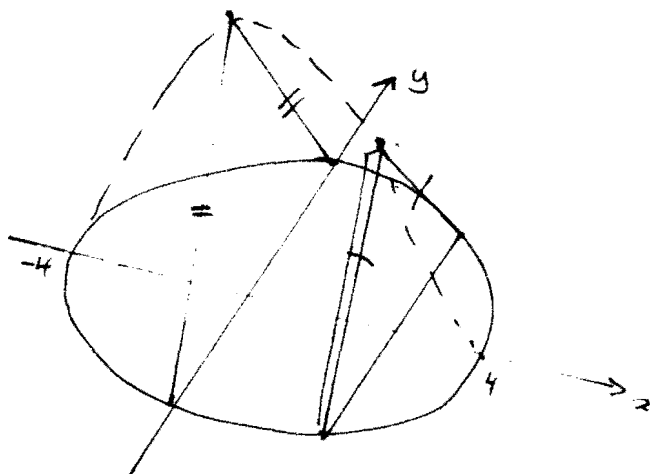
$$I_3 + I_1 = \frac{1}{2}$$

$$\begin{aligned}
 I_1 &= \int_0^{\pi/4} \tan x \cdot dx \\
 &= \left[-\ln(\cos x) \right]_0^{\pi/4} \\
 &= -\ln\left(\frac{1}{\sqrt{2}}\right) + \ln 1 \\
 &= -\ln \frac{1}{\sqrt{2}} \\
 &= \ln \sqrt{2}
 \end{aligned}$$

$$\begin{aligned}
 \therefore I_7 &= \frac{1}{6} - I_5 \\
 &= \frac{1}{6} - \left(\frac{1}{4} - I_3 \right) \\
 &= -\frac{1}{12} + I_3 \\
 &= -\frac{1}{12} + \frac{1}{2} - I_1 \\
 &= \frac{5}{12} - \ln \sqrt{2}
 \end{aligned}$$

Question 4

a)



$$(1) \Delta V = \frac{1}{2} \cdot 2y \cdot (16-x^2) \cdot \Delta x$$

$$= y(16-x^2) \cdot \Delta x$$

but $x^2 + y^2 = 16$
 $y = \sqrt{16-x^2}$
 ← 1

$$= \sqrt{16-x^2} \cdot (16-x^2) \cdot \Delta x$$

$$= (16-x^2)^{3/2} \cdot \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=-4}^4 (16-x^2)^{3/2} \cdot \Delta x$$

$$= \int_{-4}^4 (16-x^2)^{3/2} \cdot dx$$

$$= 2 \int_0^4 (16-x^2)^{3/2} \cdot dx$$

since integrand is
 on even fn. must
 mention

ii) Let $x = 4 \sin \theta$

$$dx = 4 \cos \theta \cdot d\theta$$

$$16-x^2 = 16 - 16 \sin^2 \theta = 16 \cos^2 \theta$$

If $x=0$, $\theta=0$

If $x=4$, $\sin \theta=1$, $\theta = \pi/2$

$$V = 2 \int_0^4 \sqrt{16 \cos^2 \theta}^3 \cdot 4 \cos \theta \cdot d\theta$$

$$= 2 \int_0^4 64 \cos^3 \theta \cdot 4 \cos \theta \cdot d\theta$$

$$= 512 \int_0^{\pi/2} \cos^4 \theta \cdot d\theta$$

$$\cos^4 \theta = \left(\frac{1}{2} (1 + \cos 2\theta) \right)^2$$

$$= 512 \int_0^{\pi/2} \frac{1}{4} (1 + 2\cos 2\theta + \cos^2 2\theta) \cdot d\theta \quad |$$

$$= 128 \int_0^{\pi/2} 1 + 2\cos 2\theta + \frac{1}{2} + \frac{1}{2} \cos 4\theta \cdot d\theta$$

$$= 128 \left[\frac{3}{2} \theta + \sin 2\theta + \frac{1}{8} \sin 4\theta \right]_0^{\pi/2} \quad |$$

$$= 128 \left(\left(\frac{3\pi}{4} + 0 + 0 \right) - 0 \right)$$

$$= \underline{96\pi \text{ units}^3} \quad |$$

$$b) i) \int_0^{2a} f(x) \cdot dx = \int_0^a f(x) \cdot dx + \underbrace{\int_a^{2a} f(x) \cdot dx}_{\quad |}$$

Now if $u = 2a - x$ then $x = 2a - u$, $dx = -du$

If $x = 2a$, $u = 0$

If $x = a$, $u = a$

$$\therefore \int_a^{2a} f(x) \cdot dx = \int_a^0 f(2a - u) \cdot -du \quad |$$

$$= \int_0^a f(2a - u) \cdot du$$

$$= \int_0^a f(2a - x) \cdot dx \quad |$$

$$\therefore \text{Original integral} = \int_0^a f(x) + f(2a - x) \cdot dx$$

$$ii) \int_0^{\pi} \frac{x}{2+\sin x} \cdot dx \quad \left[f(x) = \frac{x}{2+\sin x} \right]$$

$$= \int_0^{\pi/2} f(x) + f(\pi-x) \cdot dx$$

$$= \int_0^{\pi/2} \frac{x}{2+\sin x} + \frac{\pi-x}{2+\sin(\pi-x)} \cdot dx$$

$$= \int_0^{\pi/2} \frac{x}{2+\sin x} + \frac{\pi-x}{2+\sin x} \cdot dx$$

since
 $\sin(\pi-x) = \sin x$

$$= \int_0^{\pi/2} \frac{\pi}{2+\sin x} \cdot dx$$

$$= \pi \int_0^{\pi/2} \frac{1}{2+\sin x} \cdot dx$$

$$\therefore \boxed{k = \pi}$$